

Case Study: E-commerce Management (Products Collection)

Scenario

An **e-commerce company** manages its products in a MongoDB collection called **products**.

Each product contains:

- name
- category
- price
- stock
- brand

```
db.products.insertMany([
{
  name: "iPhone 14",
  category: "Electronics",
  price: 75000,
  stock: 25,
  brand: "Apple"
},
{
  name: "Samsung Galaxy S22",
  category: "Electronics",
  price: 65000,
  stock: 30,
  brand: "Samsung"
},
{
  name: "Dell Laptop",
  category: "Computers",
  price: 55000,
  stock: 15,
  brand: "Dell"
},
{
```

```
name: "HP Pavilion Laptop",
category: "Computers",
price: 50000,
stock: 12,
brand: "HP"
},
{
name: "Nike Shoes",
category: "Fashion",
price: 5000,
stock: 50,
brand: "Nike"
},
{
name: "Adidas T-shirt",
category: "Fashion",
price: 1500,
stock: 80,
brand: "Adidas"
},
{
name: "Sony Headphones",
category: "Electronics",
price: 3000,
stock: 40,
brand: "Sony"
},
{
name: "Wooden Dining Table",
category: "Furniture",
price: 20000,
stock: 5,
brand: "IKEA"
},
{
name: "Office Chair",
category: "Furniture",
price: 7000,
stock: 20,
brand: "Godrej"
}
```

```
},  
{  
  name: "Bluetooth Speaker",  
  category: "Electronics",  
  price: 2500,  
  stock: 60,  
  brand: "JBL"  
}  
])
```

Find Operations

Find all products in the collection.

Display all products in a well-formatted structure.

Find a single product using its email/contact (or unique field like product name).

Find products where category is "Electronics".

Find products where price is less than 5000.

Find products where price is less than or equal to 5000.

Find products where price is greater than 5000.

Find products where price is greater than or equal to 5000.

Find products where price is not equal to 5000.

Find products where price is in the list [1000, 2000, 5000].

Find products where price is not in the list [1000, 2000, 5000].

Find products where category is "Electronics" AND price is 5000.

Find products where category is "Electronics" OR brand is "Apple".

Find products where price is greater than 5000 OR brand is "Samsung".

Find products where price is between 5000 and 20000.

Find products where price is NOT greater than 5000.

Update Operations

Update product name using a specific product ID.

Update stock for products where price is less than 5000.

Update price for one product that meets a condition.

Update stock for multiple products where category is "Electronics".

Replace an entire product document with new details.

Delete Operations

Delete a product using its name or email.

Delete one product where price is less than 2000.

Delete all products from the collection (like truncate).

Projection & Filtering

Display only product name and price fields.

Display product name and price without showing ID.

Limit & Pagination

Display only the first 5 products.

Skip first 5 products and display next 5 products.

Sorting

Sort products by price in ascending order.

Sort products by price in descending order.